Journal of Physics D:
Applied Physics

PAPER

Numerical study on ionic wind from pin to mesh with hole configuration under DC negative corona discharge

To cite this article: Ji Hong Chung *et al* 2025 *J. Phys. D: Appl. Phys.* **58** 105205

IOP Publishing

Journal of Physics D: Applied Physics

J. Phys. D: Appl. Phys. 58 (2025) 105205 (10pp)

https://doi.org/10.1088/1361-6463/ada2a5

Numerical study on ionic wind from pin to mesh with hole configuration under DC negative corona discharge

Ji Hong Chung[1,4], **Tae Sung Ahn**[2,4], **Dong Kee Sohn**[1] **and Han Seo Ko**[1,3,*]

[1] School of Mechanical Engineering, Sungkyunkwan University, 2066, Seobu-ro, Jangan-gu, Suwon 16419, Republic of Korea

[2] Department of Semiconductor and Display Engineering, Sungkyunkwan University, 2066, Seobu-ro, Jangan-gu, Suwon 16419, Republic of Korea

[3] Department of Smart Fab. Technology, Sungkyunkwan University, 2066, Seobu-ro, Jangan-gu, Suwon 16419, Republic of Korea

E-mail: hanseoko@skku.edu

Received 21 October 2024, revised 17 December 2024

Accepted for publication 23 December 2024

Published 7 January 2025

Abstract

This study explores the enhancement of ionic wind generation efficiency by incorporating a central hole in the mesh electrode under DC negative corona discharge conditions. Using numerical analysis techniques such as drift-diffusion equations and the Poisson equation, the mechanisms driving ionic wind generation were extensively analyzed. The characteristic Trichel pulses of negative corona discharge were examined by monitoring the variations in current with changes in applied voltage and central hole diameter. Particle image velocimetry experiments validated numerical analysis, showing good agreement between experimental and numerical results. The introduction of a central hole in the mesh electrode significantly reduced pressure drop and increased ionic wind velocity, addressing common problems of decreased momentum and increased frictional loss in mesh electrodes. This modification allows the ionic wind to flow more freely through the mesh electrode, enhancing overall efficiency by mitigating energy losses typically associated with mesh structures. Additionally, the study found that ionic wind velocity increased with rising voltage, showing variations in maximum velocity depending on hole size. The results suggest that optimizing central hole size could enhance the efficiency of ionic wind generation devices in various applications.

Keywords: negative corona discharge, numerical simulation, ionic wind, mesh electrode, central hole, Trichel pulse

1. Introduction

Various devices designed to generate fluid flow have been developed, among which electrohydrodynamic (EHD) pumps, based on corona discharge, have emerged as prominent over recent decades. EHD pumps or ionic wind generators can convert electrical energy into kinetic energy, thereby producing ionic wind. These devices are characterized by quiet operation, low power consumption, rapid response, and the ability to generate flow on demand without moving mechanical parts. Thus, they have found applications in diverse fields such as thermal management, food drying, dust removal, and electric propulsion [1–7].

Ionic wind, resulting from the corona discharge of gases at atmospheric pressure, occurs when strong voltage is applied to sharp pins or wires, ionizing air into positive ions, negative ions, and electrons. Foundational studies on corona discharge, including Peek, Loeb and others, have established

[4] These authors are contributed equally to this work.

[*] Author to whom any correspondence should be addressed.

these principles and laid the groundwork for ongoing research on ionic wind [8–11]. Ionized particles, accelerated by the electric field, collide with neutral air, transferring momentum. The region where ionization occurs is defined as the ionization zone, while the region where momentum energy transfer through collisions occurs is termed the drift zone, generating a flow known as ionic wind. More recent studies have advanced the understanding of mass and species transport within corona discharges [12–17].

Ionic wind technology is advanced across various implementations, differentiated by electrode types, configurations, and the magnitude of voltage and current. Typical methods include pin-plate, pin-ring, pin-mesh, wire-plate, wire-ring, and wire-mesh configurations. These devices vary in size, efficiency, and application. Additionally, several studies investigate multi-electrode configurations, yet an optimized shape remains to be determined. Recent research, supported by experimental validation, focuses on probing electrode structures and operating conditions to maximize flow generation [18, 19]. Among these, the pin-mesh type is widely favored as an ionic wind generator due to its high ionization efficiency, low operating voltage, low energy consumption, and a simple, reliable structure, as substantiated by extensive research [20–24]. However, in this system, the mesh electrode functions as both the collector electrode and the outlet, which results in significant frictional losses due to inherent structural limitations and generally lower ionic wind velocity. Previous studies, such as those by Zeng *et al* and Iranshahi *et al*, have analyzed ionic wind generation efficiency in relation to mesh parameters including wire diameter, number of wires, and mesh open area [25, 26]. While these efforts focused on optimizing mesh characteristics, challenges to reduce frictional loss and further enhance ionic wind velocity persist. Traditional modifications have not fully resolved these problems, creating a demand for innovative approaches, such as the central hole modification explored in this study.

Trichel pulses are distinctive periodic features specific to negative corona discharge, characterized by regular oscillations in current that arise from complex interactions between ionization and recombination processes near the electrode [27, 28]. Experimental studies, such as Lama and Gallo, have examined Trichel pulse dependence of parameters like pulse frequency, charge per pulse, and current in relation to applied voltage and electrode spacing [29, 30]. These pulses in negative corona discharge are generated through interactions among multiple charged species. Therefore, accurately modeling negative corona discharge and ionic wind necessitates a comprehensive understanding of the intricate relationships involving electrical, physical, and chemical reactions. Simplifying and interpreting these reactions is vital due to the complex nature of corona discharge, which encompasses various chemical processes. Peek's earlier work unveiled chemical reactions between oxygen and nitrogen during corona discharge, resulting in phenomena such as sound, light, and heat [8]. Peek also provided equations for calculating the electric field strength necessary to initiate corona discharge, considering factors like air density and electrode radius.

Advancements in industrial and research applications have driven a surge in interest in corona discharge analysis, leading to the adoption of methods that utilize Poisson's equation and drift-diffusion equations for calculating electric field and spatial charge density [31]. One such method, unipolar analysis, relies on the Kaptzov hypothesis and neglects the ionization zone, positing that the electrode's electric field remains consistent once the initiation voltage is surpassed [32]. Morrow enhanced these techniques by integrating three gas reactions and combining Poisson's equation with three drift-diffusion equations for positive ions, negative ions, and electrons [33]. This methodology has been employed in various studies, including analyses comparing negative and positive corona discharges [34]. Additionally, Trichel pulses were explored using this integrated method [35, 36]. Hagelaar developed a method to calculate electron transport and reaction coefficients using the Boltzmann equation based on cross-sectional data, subsequently releasing the open simulator BOLSIG+ [37]. Current research encompasses over twenty reactions occurring in corona discharge in air and extends to studies involving gases such as CO_2, SF_6, and Ar [38–40].

This study focused on enhancing ionic wind velocity by introducing a hole in the central region of the mesh electrode. Introducing a hole in the center of the mesh electrode reduces pressure drop; however, as the hole size increases, ionization efficiency declines, resulting in diminished flow velocities [41]. A numerical analysis was performed to optimize ionic wind velocity based on the hole size. The validity of this numerical analysis was confirmed through particle image velocimetry (PIV) experiments and time-averaged current measurements. For DC negative corona discharge, numerical analysis focusing on three reactions of positive ions, negative ions, and electrons was conducted to form Trichel pulses. EHD forces derived from corona discharge analysis were employed to represent ionic wind flow.

2. Numerical methods

In this study, numerical analysis using the commercial software 'COMSOL Multiphysics', was conducted. A 2D axisymmetric model was utilized to represent the Trichel pulse in negative corona discharge, accounting for three species of positive ions, negative ions, and electrons.

Key parameters such as the pin tip radius and the mesh electrode distance are crucial for corona discharge. For the analysis, a pin tip radius of 0.05 mm and an electrode distance of 20 mm were used, and the central hole diameters in the mesh electrode, ranging from 0 to 20 mm, were investigated. Experimental observations showed corona discharge initiation at applied voltages ranging from −5 to −6 kV, with sparks occurring between −14 and −16 kV. Thus, the applied voltage was examined in the range of −6 to −14 kV.

A mesh electrode with a wire diameter of 0.18 mm and a hole diameter of 0.23 mm was employed. Although a 3D model would provide a more detailed representation of the experiments, the complexity and computational time required the assumption of a planar mesh electrode and 2D analysis, treating it as a porous medium.

2.1. Governing equation of corona discharge and swarm parameter

Using the Kaptzov hypothesis and Peek`s equations enables the calculation of spatial charge distribution in corona discharge [42–45]. This approach, however, does not account for the interactions among positive ions, negative ions, and electrons, and thus fails to accurately represent Trichel pulses. Additionally, the multitude of ionization reactions occurring in air during corona discharge poses a challenge, as analyzing all these reactions would be excessively time-consuming and inefficient for optimization purposes. Therefore, to represent Trichel pulses while minimizing computational time, we considered relatively simplified interactions among positive ions, negative ions, and electrons. Drift-diffusion equations for each species and Poisson`s equation for the electric field were resolved as follows:

$$\frac{\partial n_{\mathrm{p}}}{\partial t}+\nabla\cdot\left(\mu_{\mathrm{p}}\vec{E}n_{\mathrm{p}}-D_{\mathrm{p}}\nabla n_{\mathrm{p}}\right)=\alpha n_{\mathrm{e}}\left|\mu_{\mathrm{e}}\vec{E}\right|-k_{\mathrm{ep}}n_{\mathrm{e}}n_{\mathrm{p}}-k_{\mathrm{np}}n_{\mathrm{n}}n_{\mathrm{p}} \tag{1}$$

$$\frac{\partial n_{\mathrm{n}}}{\partial t}+\nabla\cdot\left(-\mu_{\mathrm{n}}\vec{E}n_{\mathrm{n}}-D_{\mathrm{n}}\nabla n_{\mathrm{n}}\right)=\eta n_{\mathrm{n}}\left|\mu_{\mathrm{n}}\vec{E}\right|-k_{\mathrm{np}}n_{\mathrm{n}}n_{\mathrm{p}} \tag{2}$$

$$\frac{\partial n_{\mathrm{e}}}{\partial t}+\nabla\cdot\left(-\mu_{\mathrm{e}}\vec{E}n_{\mathrm{e}}-D_{\mathrm{e}}\nabla n_{\mathrm{e}}\right)=\alpha n_{\mathrm{e}}\left|\mu_{\mathrm{e}}\vec{E}\right|-\eta n_{\mathrm{e}}\left|\mu_{\mathrm{e}}\vec{E}\right|-k_{\mathrm{ep}}n_{\mathrm{e}}n_{\mathrm{p}} \tag{3}$$

$$\nabla\cdot\vec{E}=-\nabla^{2}V=\frac{e\left(n_{\mathrm{p}}-n_{\mathrm{e}}-n_{\mathrm{n}}\right)}{\varepsilon_{0}\varepsilon_{\mathrm{r}}} \tag{4}$$

where $n_{\mathrm{e}}, n_{\mathrm{p}}$, n_{n}, $\mu_{\mathrm{e}}, \mu_{\mathrm{p}}$, μ_{n}, D_{e}, D_{p} and D_{n} are the number density, mobility and diffusion coefficients for electrons, positive ions and negative ions, respectively. And $\vec{E}$, V, e, ε_0, and ε_{r} denote the electric vector, electric potential, elementary charge, vacuum permittivity, and relative permittivity, respectively. The main reactions include ionization α, attachment through two-body and three-body collisions η, recombination coefficients of positive ions with electrons k_{ep} and recombination coefficients of positive ions with negative ions k_{np}. The mobilities of electrons, ionization, and attachment rates can be expressed as functions of the electric field. The swarm parameter used in this study is summarized in table 1 [46, 47].

In corona discharge, electrons, accelerated by a strong electric field near sharp electrodes, gain sufficient energy to collide with neutral particles, causing ionization and generating positive ions and electrons. Conversely, in regions with weak electric fields, electrons lack the energy needed for ionization, attach to neutral particles, and primarily form negative ions. Additionally, recombination occurs when electrons and positive ions collide with negative ions, leading to the formation of neutral particles. These reactions are fundamental to the formation of corona discharge. The representative chemical reactions of corona discharge at atmospheric pressure include:

Table 1. Swam parameters [46, 47].

Swarm parameter	Value	Units
μ_{e} [46]	$\frac{3.74\times10^{22}}{\mathrm{N}}\left(\frac{\lvert\vec{E}\rvert}{\mathrm{N}}\right)^{-0.25}$	$\mathrm{cm^2V^{-1}s^{-1}}$
μ_{p} [47]	2.43	
μ_{n} [47]	2.7	$\mathrm{cm^2V^{-1}s^{-1}}$
α [46]	$N\times1.4\times10^{-16}\exp\left(-\frac{600}{\lvert\vec{E}\rvert/N}\right)$	
η_2 [46]	$N\times6\times10^{-19}\exp\left(-\frac{100}{\lvert\vec{E}\rvert/N}\right)$	$\mathrm{cm^{-1}}$
η_3 [46]	$N^2\times1.6\times10^{-37}\left(\frac{\lvert\vec{E}\rvert}{\mathrm{N}}\right)^{1.1}$	$\mathrm{cm^{-1}}$
k_{ep} [46]	5×10^{-8}	$\mathrm{cm^3s^{-1}}$
k_{np} [46]	$2\times10^{-6}(T/300)^{-1.5}$	$\mathrm{cm^3s^{-1}}$
D_{e} [46]	$1800\times760/p$	$\mathrm{cm^2s^{-1}}$
D_{p} [47]	0.028	$\mathrm{cm^2s^{-1}}$
D_{n} [47]	0.043	$\mathrm{cm^2s^{-1}}$

$$\mathrm{e}+\mathrm{N_2 \ or\ O_2}\rightarrow 2\mathrm{e}+\mathrm{N_2^+\ or\ O_2^+}\,\mathrm{Ionization} \tag{5}$$

$$\mathrm{e}+\mathrm{O_2}\rightarrow\mathrm{O^-}+\mathrm{O},\ \mathrm{e}+\mathrm{O_2}+\mathrm{O_2}\rightarrow\mathrm{O_2^-}+\mathrm{O_2}\mathrm{Attachment} \tag{6}$$

$$\mathrm{e}+N_4^+\ \mathrm{or\ O_4^+}\rightarrow 2\mathrm{N_2\ or\ 2O_2}\mathrm{Recombination\ with\ Positive\ Ions\ and\ Electrons} \tag{7}$$

$$\mathrm{O_2^-}+\mathrm{O_2^+}+\mathrm{N_2\ or\ O_2}\rightarrow 2\mathrm{O_2}+\mathrm{N_2\ or\ O_2}\mathrm{Recombination\ with\ Positive\ and\ Negative\ Ions}\ . \tag{8}$$

The corona discharge current was computed using the formula proposed by Sato [48]. Equation (9), derived from the principle of energy conservation between two electrodes, calculates the displacement current resulting from the movement of charged particles under a constant electric field

$$I=\frac{e}{V}\iint 2\pi r\left(\Gamma_{\mathrm{p}}-\Gamma_{\mathrm{n}}-\Gamma_{\mathrm{e}}\right)\overrightarrow{E_{\mathrm{L}}}\mathrm{d}r\mathrm{d}z \tag{9}$$

where, Γ_{p}, Γ_{n}, and Γ_{e} represent the electric flux of positive ions, negative ions, and electrons, respectively, while r and z represent the cylindrical coordinates.

2.2. Governing equation of fluid dynamics

In negative corona discharge, positive ions move toward the discharge electrode, while negative ions and electrons move toward the ground. This process bifurcates into an ionization region, where high electric fields cause ionization, and a drift region, where electrons and negative ions move toward the ground. In the ionization region, positive ions collide with the negative electrode, emitting secondary electrons, whereas in the drift region, electrons and negative ions moving toward the ground collide with neutral particles. These collisions promote momentum transfer, inducing airflow known as ionic wind. The resultant force is referred to as the EHD force, defined by equation (10) [49]

$$F_{\mathrm{EHD}} = e\left(n_{\mathrm{p}} - n_{\mathrm{n}} - n_{\mathrm{e}}\right)\vec{E}. \tag{10}$$

The corona discharge model and the fluid model are coupled unidirectionally, indicating that the fluid model minimally impacts the corona discharge model [50]. The time scale of the corona discharge model is in the microsecond range, whereas the fluid model operates in the millisecond range. To reconcile the time scale disparity between the two models, the EHD force was calculated by averaging over time. The numerical analysis of ionic wind employed the continuity equation and Navier–Stokes equation as foundational models, represented as follows:

$$\nabla\left(\rho_{\mathrm{g}} u\right) = 0 \tag{11}$$

$$\rho_{\mathrm{g}}\left(u\cdot\nabla\right)u = -\nabla P + \nabla\left(\left(\mu+\mu_{\mathrm{t}}\right)\left(\nabla u + \left(\nabla u\right)^{T}\right)\right) + F_{\mathrm{EHD}} + \rho_{\mathrm{g}} g \tag{12}$$

where, ρ_{g}, u, P, μ, μ_{t}, and g represent the density of air, air velocity, absolute pressure, air dynamic viscosity, turbulent dynamic viscosity, and gravitational acceleration, respectively. Numerical analysis of the mesh electrode requires significant computational resources. To improve computational efficiency, the mesh electrode was modeled as a porous medium. The Navier–Stokes equation, which accounts for the pressure drop due to porous media, is expressed as follows:

$$\rho_{\mathrm{g}}\left(\mathrm{u}\cdot\nabla\right)\mathrm{u} = -\nabla P + \nabla\left(\mu\frac{1}{\epsilon_{\mathrm{p}}}\left(\nabla u + \left(\nabla u\right)^{T}\right) - \frac{2}{3}\mu\frac{1}{\epsilon_{\mathrm{p}}}\left(\nabla\cdot u\right)\right) + \left(-\left(\mu\kappa^{-1} + \frac{c_{\mathrm{F}}}{\sqrt{\kappa}}\rho_{\mathrm{g}}\left|u\right| + \frac{\mathrm{Q_m}}{\epsilon_{\mathrm{p}}^{2}}\right)\mathrm{u}\right) + F_{\mathrm{EHD}} + \rho_{\mathrm{g}} g \tag{13}$$

where, ϵ_{p}, κ and c_{F} represent the porosity (0.4204), permeability (1.49×10^{-9} m^2), and the Forchheimer coefficient (0.000 16), respectively. The Forchheimer coefficient accounts for the non-linear drag effects in porous media flow at higher velocities. The assumption of porous media was validated by comparing the pressure drop using 3D fluid analysis, which resulted in a minimal error of 2%.

2.3. Boundary conditions and grid

The boundary conditions used are summarized in figure 1 and table 2. The pin electrode is subjected to an applied voltage, while the mesh electrode is grounded. In the experimental setup, the wall exhibits a dielectric boundary condition with a relative permittivity of 2.8 and a thickness of 5 mm. Open space is assumed to be charge-free. For the drift-diffusion equations, ions and electrons exhibit either outflow or zero density at the wall and open space. Electron emission is modeled as a boundary condition at the pin electrode, with

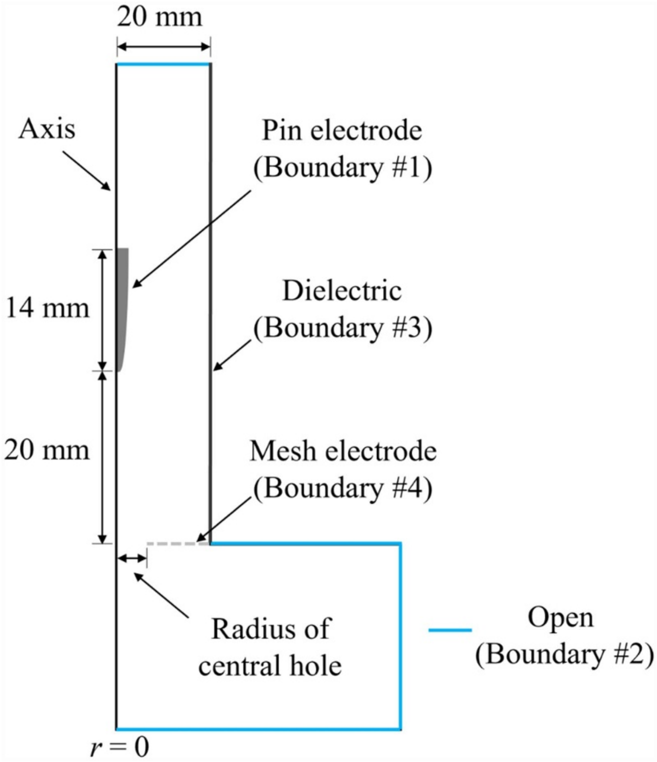

Figure 1. Geometry and boundary conditions.

the secondary electron emission coefficient (SEEC, γ) set at 0.0004 [47]. The electron flux applied to the pin electrode is as follows:

$$\Gamma_{\mathrm{e}} = \gamma n_{\mathrm{p}}\mu_{\mathrm{p}}\left|\vec{E}\right| \tag{14}$$

To depict the characteristics of the quasi-neutral plasma, the initial condition for negative ions was set to zero, while a Gaussian distribution was used for the initial conditions of electrons and positive ions, defined as follows [51]:

$$n_{\mathrm{e,p}}\left(t=0\right) = 10^{16}\exp\left(-\frac{r^{2}}{2*0.025^{2}} - \frac{z^{2}}{2*0.025^{2}}\right). \tag{15}$$

Ionization and attachment reactions occur locally near the pin electrode tip. In this region, where electrons, positive ions, and negative ions move at very high velocities, grid resolution plays a crucial role. Grid sizes at the pin electrode tip were set to a minimum of 0.0001 mm, with a size of 0.001 mm within a 0.2 mm radius from the tip. The grid size near the mesh electrode was set to 0.01 mm. All grids were triangular, comprised of 320 000 grid points, and had a grid quality ranging from 0.08 to 0.45, which was considered satisfactory.

The frequency of Trichel pulses typically ranges from approximately 100 kHz to 2 MHz. Thus, selecting an appropriate time step is essential for accurately simulating Trichel pulses. The initial time step was set very low, at 0.001 ns, to prevent divergence in early stages of iterations. The maximum time step was set to 10 ns, and simulations were conducted up

Table 2. Boundary conditions.

Parameter	Boundary #1	Boundary #2	Boundary #3	Boundary #4
Electron	Equation (14) SEEC	Outflow $-n \cdot (-D_e \nabla n_e) = 0$	Outflow $-n \cdot (-D_e \nabla n_e) = 0$	Outflow $-n \cdot (-D_e \nabla n_e) = 0$
Positive ion	Outflow $-n \cdot (-D_p \nabla n_p) = 0$	Outflow $-n \cdot (-D_p \nabla n_p) = 0$	Outflow $-n \cdot (-D_p \nabla n_p) = 0$	Zero density $n_p = 0$
Negative ion	Zero density $n_n = 0$	Outflow $-n \cdot (-D_n \nabla n_n) = 0$	Outflow $-n \cdot (-D_n \nabla n_n) = 0$	Outflow $-n \cdot (-D_n \nabla n_n) = 0$
Electric potential	$V = V_0$	$n \cdot D = 0$	$n \cdot (D_1 - D_2) = -\nabla_T \cdot d_s (\varepsilon_0 \varepsilon_r \nabla_T V)$	$V = 0$

to 0.000 05 s. After simulating Trichel pulses until 0.0005 s, time-averaged values were used to analyze the EHD force.

3. Experiments and validations

3.1. Validation of corona discharge model

Accurate simulation of corona discharge requires extremely small grid sizes. However, conducting a 3D simulation for the mesh electrode incurs significant computational costs. Consequently, the mesh electrode was simplified to a planar configuration, and its validity was assessed. In atmospheric pressure corona discharge, electron collision frequency is exceptionally high, resulting in most parameters being expressed as functions of the electric field. Therefore, the electric field is crucial in simulating corona discharge. A comparative analysis was carried out between the 3D electric field of the mesh electrode and the planar electrode. The analysis utilized the Laplace equation:

$$\nabla \cdot \vec{E} = -\nabla^2 V = 0. \tag{16}$$

The maximum electric field at the tip was found to be 6.57×10^7 V m^{-1} for the planar configuration and 6.59×10^7 V m^{-1} for the mesh electrode, indicating a small discrepancy of 0.3%. Additionally, when comparing the radial electric field at the ground, a discrepancy of 2% was observed, as illustrated in figure 2. Given the negligible discrepancy and the impracticality of ionization reactions at the mesh electrode, assuming the mesh electrode as planar is deemed valid.

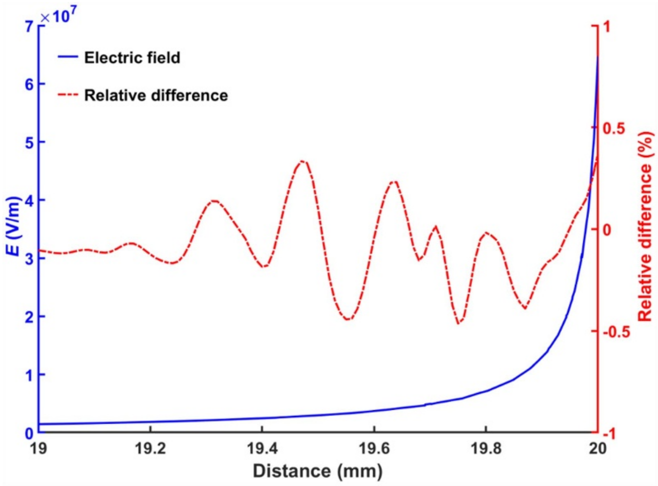

Figure 2. Electric field and relative difference near the pin electrode.

3.2. Validation of ionic wind model

The validation of the ionic wind model was achieved by comparing the momentum transfer and induced ionic wind phenomena with the results of PIV experiments. The PIV and current measurements were conducted five times using the setup shown in figure 3. Di-ehtlyhexyl-sebacate particles were sufficiently injected into the chamber, and a negative voltage was applied to the pin electrode generating ionic wind. The particle motion was tracked using a continuous laser and a high-speed camera, allowing the distribution of the generated ionic wind to be calculated. The standard deviation of the measured ionic wind velocity across the repeated experiments was approximately 5%. Current was determined based on the obtained values from a passive voltage probe. The comparison of velocity took place 20 mm from the mesh electrode, where it was found that ions and electrons had diminished by more than 99.9%. When scenarios involving a mesh electrode with and without a central hole were analyzed, velocities were generally lower in the absence of a central hole, primarily due to pressure drop, as shown in figure 4.

This reduction underscored a significant velocity decrease at the mesh electrode, highlighting the need for enhancement. In contrast, configurations with a central hole displayed significantly increased velocities at the center, with maximum velocities reaching roughly five times higher. Both simulation and experimental results demonstrated the characteristic centralized flow of ionic wind, indicating a good agreement between the two. This alignment confirmed the successful replication of ionic wind characteristics. Additionally, comparison of average velocity showed a minor error of 4.8%, further substantiating the accuracy of the numerical analysis results.

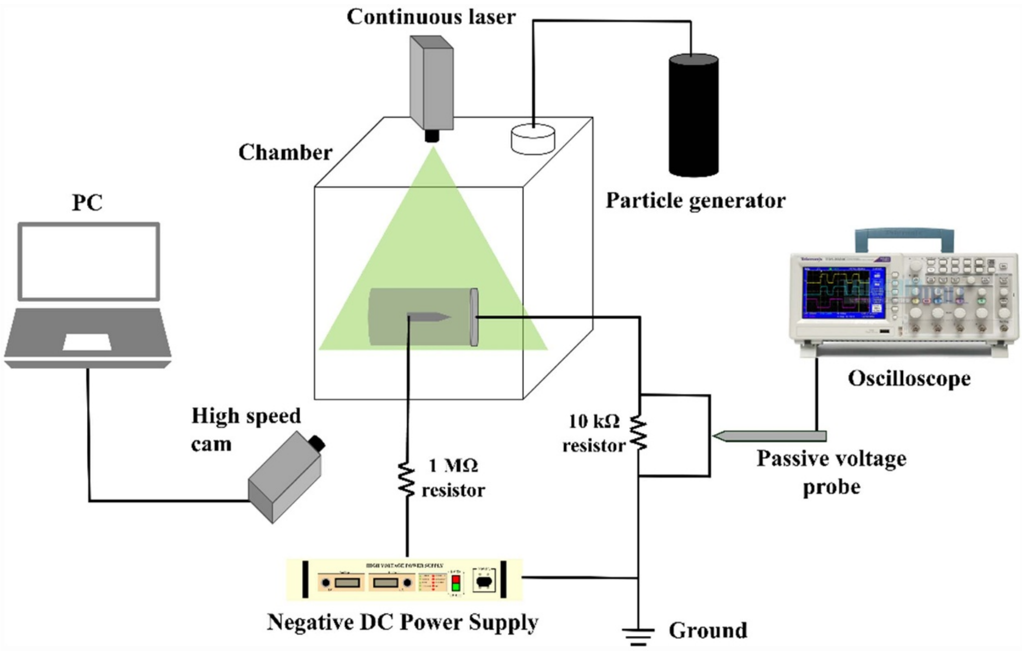

Figure 3. Experimental setup for PIV.

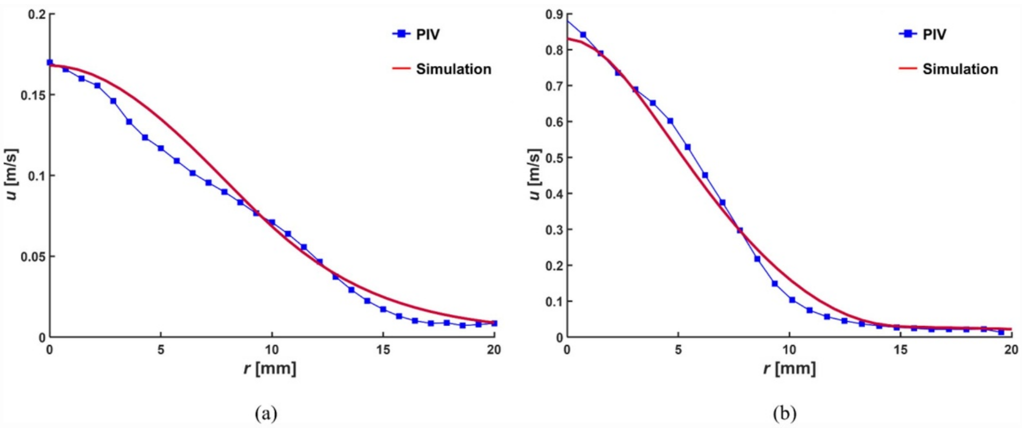

Figure 4. Velocity distribution at −6 kV; (a) without a central hole and (b) with a central hole of 10 mm.

4. Results and discussion

4.1. Trichel pulse

The negative corona discharge is characterized by the generation of Trichel pulses, as depicted in figure 5. Figure 6 illustrates the mechanism underlying Trichel pulse generation, divided into four distinct steps. Step 1 initiates the preparatory stage, where clouds of negative ions, positive ions, and electrons are formed near the pin. Concurrently, positive ions attach to the tip, facilitating further ion cloud formation. In this phase, a relatively weak electric field is formed near the pin, which, along with the generation of electrons and negative ions, promotes more ion generation than annihilation.

Step 2 is characterized by the peak current phase of the Trichel pulse, where the maximum densities of negative ions, positive ions, and electrons are presented, accompanied by a corresponding peak in the electric field induced by charged particles. The acceleration of positive ion annihilation exceeds the ionization reaction rate due to the increased electric field near the pin, resulting in ionization suppression and current reduction. Meanwhile, the increased positive ion density facilitates the acceleration of negative ions and electrons toward the ground electrode.

Step 3 represents the acceleration phase of the negative ion cloud, during which maximum positive and negative ions are generated due to adequate ionization and attachment reactions at peak current. As the annihilation of positive ions intensifies near the pin, the negative ion cloud is propelled towards the ground electrode. While electrons also accelerate, their density rapidly decreases due to high velocities and reactions, thereby facilitating momentum transfer and the formation of ionic wind through acceleration and collision with neutral particles.

Step 4 entails preparing for the subsequent pulse, wherein negative ions are transported via acceleration and diffusion as

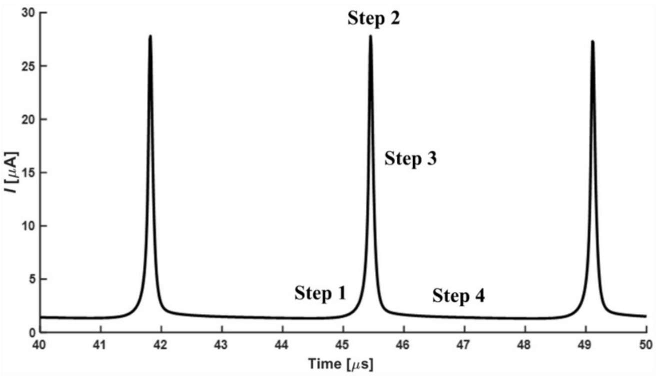

Figure 5. Trichel pulses at −6 kV.

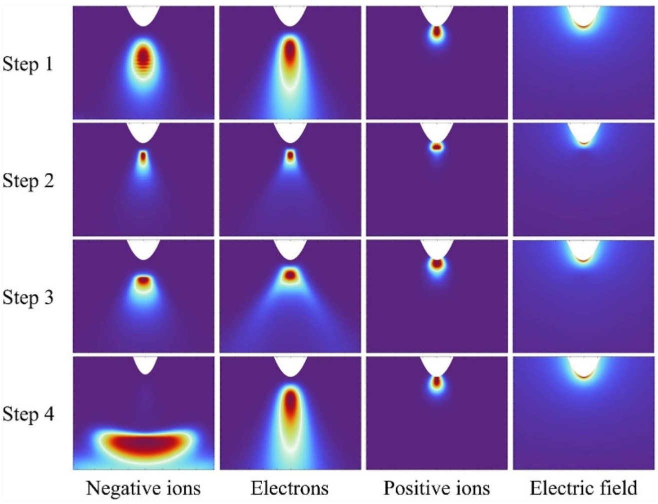

Figure 6. Steps of Trichel pulses for negative ions, electrons, positive ions, and electric field.

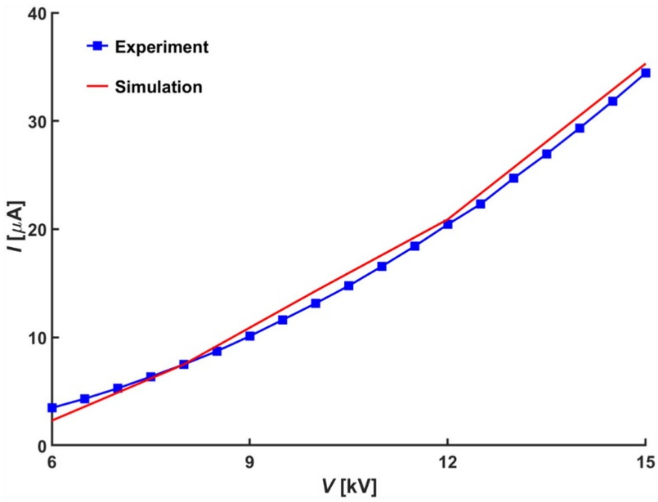

Figure 7. Time-averaged current for various applied voltages.

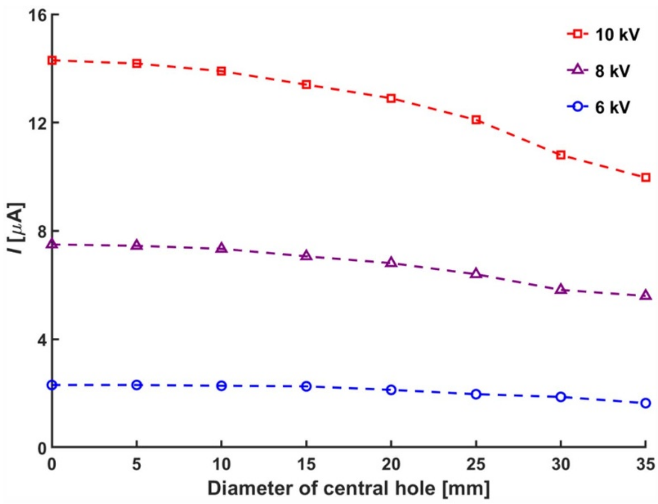

Figure 8. Current for various diameters of central hole.

dictated by the electric field, leading to a reduction in the density of the negative ion cloud. All parameters, including positive ions, negative ions, electrons, and the electric field, reach their minimum values. Near the pin, ionization reactions gradually accumulate ions and electrons in preparation for the next pulse. These four steps continually repeat to generate Trichel pulses.

The impact of applied voltage on Trichel pulse frequency and peak current was assessed, and the experimentally obtained time-averaged current was compared. With an increase in applied voltage and electric field strength, ionization and attachment reactions are enhanced, leading to higher Trichel pulse frequency and peak currents, which is consistent with trends reported in previous studies [50–52]. Figure 7 compares the time-averaged current across different applied voltages between simulation results conducted on a mesh electrode without a central hole and experimental results, demonstrating an error of 5%.

4.2. Effect of central hole on current and ionic wind

The effect of the diameter of the central hole in the mesh electrode was evaluated, considering its impact on the distance between the pin electrode and the mesh electrode. As the diameter of the central hole increases, so does this distance, which consequently reduces the electric field, ionization rate, and current. A decrease in current was observed with an increase in the diameter of the central hole, as depicted in figure 8. Specifically, when the diameter of the central hole was 10 mm, only a minor decrease in current of 2.6% was observed compared to the scenario without a central hole. However, for diameters larger than 10 mm, both the current and electric field decrease significantly.

This reduction in current implies a diminished momentum transfer in the drift region due to lower ionization rates, resulting in weaker EHD forces as calculated using equation (10). Although increasing the diameter of the central hole can alleviate pressure drop, it also leads to reduced current and weakened EHD forces. Therefore, optimizing the diameter of the central hole is essential to achieve higher velocities of the ionic wind.

The impact of the central hole diameter and applied voltage on ionic wind velocity was explored. Figure 9 illustrates the

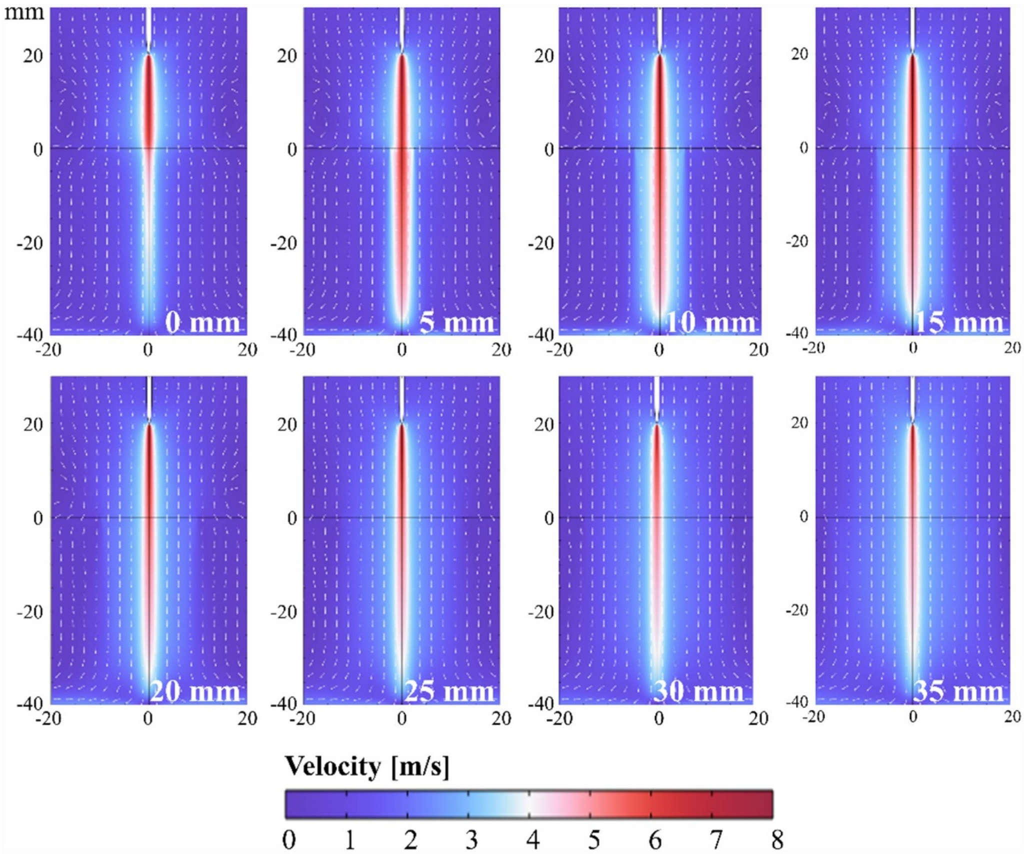

Figure 9. Velocity distribution for various diameters of central hole at applied voltage of −14 kV.

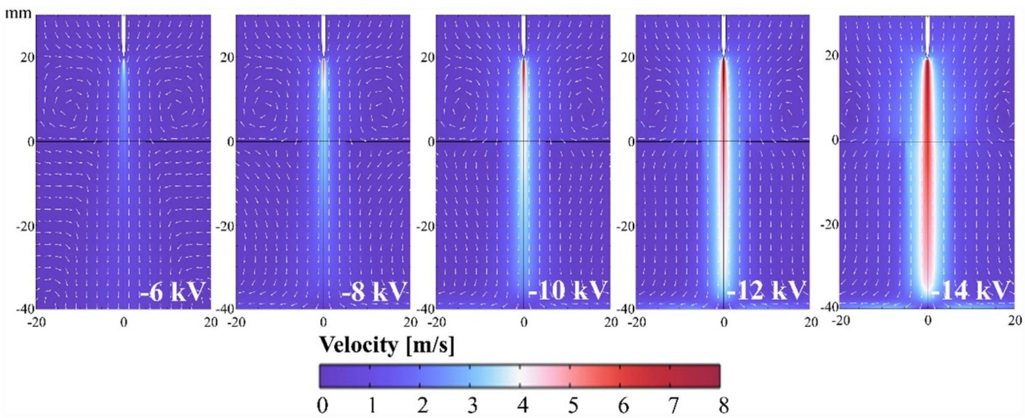

Figure 10. Velocity distribution for various applied voltages with a diameter of 10 mm.

velocity distribution for various diameters of the central hole. The current electrode configuration promotes the formation of a centralized ionic wind, where larger central holes result in increased diffusion influenced by the electric field. Although the absence of a central hole results in the highest current and the largest EHD force near the pin electrode, there is a notable reduction in ionic wind velocity due to a pressure drop as it passes through the mesh electrode. Additionally, vortices near the sidewall are observed. With an increased diameter of the central hole, the vortices decrease, accompanied by a reduction in both pressure and velocity.

Figure 10 presents the velocity distribution for various applied voltages with a diameter of 10 mm. At lower applied voltages, larger vortices form near the side wall, while at higher applied voltages, the increased momentum and inertia of the ionic wind reduce vortex formation as it penetrates the mesh electrode more directly.

With a larger central hole diameter, there is less pressure drop, leading to a smaller reduction in ionic wind velocity. However, this also increases the gap between the pin electrode and the mesh electrode, thus diminishing the EHD force. Consequently, the diameter that generates the highest

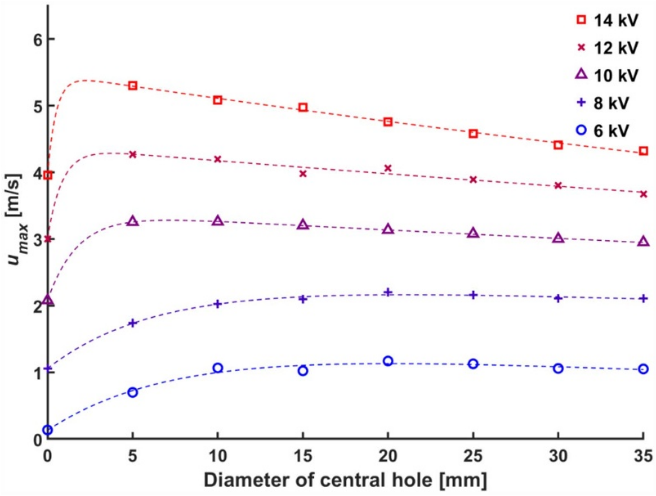

Figure 11. Maximum velocity with diameter of central hole for various applied voltages.

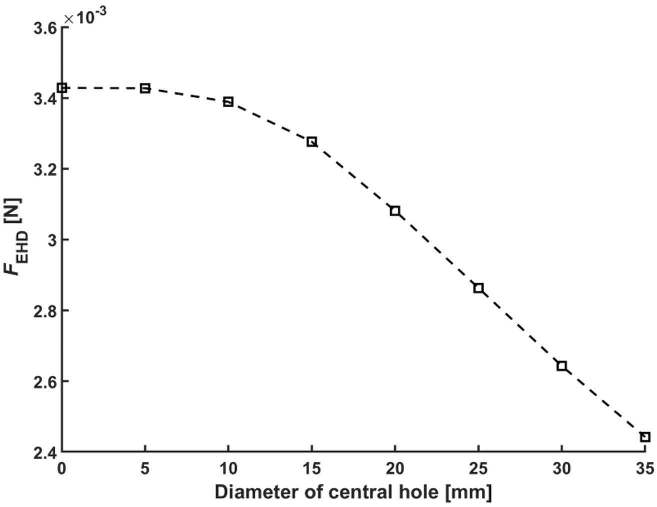

Figure 12. EHD force according to the diameter of central hole.

maximum velocity varies with each applied voltage. Figure 11 shows the maximum velocity of the ionic wind under varying central hole diameters and applied voltages. In the voltage range of −6 to −8 kV, characterized by a relatively weak EHD force and significant influence of slow velocity and diffusion, the diameter of the central hole yielding the highest maximum velocity was 20 mm due to the broader impact of the pressure drop. Conversely, in the voltage range of −10 to −14 kV, where the EHD force is stronger, the highest maximum velocity was observed with a diameter of 5 mm. This variation is attributed to a more pronounced pressure drop compared to the decrease in EHD force up to a diameter of 10 mm, as analyzed in figure 12, which illustrates how EHD force varies with the diameter of the central hole.

The optimal diameter of the central hole may vary depending on factors such as the desired velocity of the ionic wind, the distance between electrodes, and the applicable power supply. While the numerical approach and simulation results presented in this study provide valuable insights for optimizing ionic wind generation, the findings are limited to a single electrode spacing configuration, which may not fully represent all practical scenarios. Additionally, this study has not yet been extended to real-world applications, where factors such as manufacturing tolerances and environmental conditions could influence performance. Future work could explore a wider range of electrode configurations and validate the model under specific conditions.

5. Conclusion and future work

In this study, we thoroughly investigated the impact of central hole diameter on ionic wind generation using numerical analysis. By employing the drift-diffusion equation and the Poisson equation, we effectively interpreted corona discharge, a crucial mechanism in ionic wind generation. The analysis disclosed characteristic Trichel pulses of negative corona discharge and examined current variations with changes in applied voltage and central hole diameter. The numerical analysis was corroborated through PIV experiments, confirming a strong correlation between experimental and numerical results. Introducing a central hole in the mesh electrode significantly reduced the pressure drop and enhanced the ionic wind velocity. Additionally, the velocity of the ionic wind increased with higher voltages, and the maximum velocity varied according to the hole size. At −14 kV, the mesh electrode with a 5 mm central hole exhibited the highest velocity of 5.3 m s^{-1}, representing a 34% increase compared to those without a central hole, while also requiring 3% less current. Using a mesh electrode with a central hole, along with optimizing the sizes of these holes based on specific objectives, promises substantial improvements in the efficiency of ionic wind generation. These findings demonstrate the potential for enhancing ionic wind devices in practical applications such as thermal management, air purification, and propulsion systems, where improved efficiency and reduced energy consumption are critical.

By confirming that the introduction of a central hole improves ionic wind generation, further studies could explore the effects of varying electrode spacing to enhance performance under different operational conditions. Moreover, thermal management experiments will be conducted for high power electronic devices to evaluate the cooling efficiency of the proposed electrode configuration. Its energy efficiency will also be compared under equivalent conditions with other electrode configurations, such as the pin-ring type, to identify optimal designs for specific applications.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgment

This work was supported by the National Research Foundation, Korea (NRF) under Grant [RS-2024-00337022].

ORCID iDs

Ji Hong Chung https://orcid.org/0009-0005-3792-5159
Tae Sung Ahn https://orcid.org/0009-0000-8905-1231
Dong Kee Sohn https://orcid.org/0000-0001-8001-6210
Han Seo Ko https://orcid.org/0000-0003-3732-1046

References

[1] Qu J, Zeng M, Zhang D, Yang D, Wu X, Ren Q and Zhang J 2022 *J. Phys. D: Appl. Phys.* **55** 153002
[2] Wang T-H, Zhao H, You M-Y, Yan W-M and Wang S-X 2021 *Int. J. Therm. Sci.* **169** 107087
[3] Tsui Y-Y, Huang Y-X, Lan C-C and Wang C-C 2017 *J. Electrost.* **87** 1–10
[4] Dinani S T, Hamdami N, Shahedi M and Havet M 2015 *Food Bioprod. Process.* **94** 572–80
[5] Chung J H, Kim K H, Sohn D K and Ko H S 2024 *Powder Technol.* **443** 119865
[6] Mizeraczyk J, Berendt A and Podlinski J 2016 *J. Phys. D: Appl. Phys.* **49** 205203
[7] Xu H, Gomez-Vega N, Agrawal D R and Barrett S R 2020 *J. Phys. D: Appl. Phys.* **53** 025202
[8] Peek F W 1915 *Dielectric Phenomena in High Voltage Engineering* (McGrow-Hill Press)
[9] Robinson M 1962 *Am. J. Phys.* **30** 366–72
[10] Loeb L B 1965 *Electrical Coronas* (University of California Press)
[11] Christenson E A and Moller P S 1967 *AIAA J.* **5** 1768
[12] Sigmond R S and Lågstad I H 1993 *High Temp. Chem. Proces.* **2** 221–9
[13] Bahder T and Fazi C 2003 *Force on an Asymmetric Capacitor* (Army Research Laboratory)
[14] Buehler D R 2004 *J. Space Mixing* **2** 1–22 (available at: www.space-mixing-theory.com/abstract2.htm)
[15] Moreau E 2007 *J. Phys. D: Appl. Phys.* **40** 605
[16] Kiousis K N, Moronis A X and Fruh W G 2014 *Plasma Sci. Technol.* **16** 363
[17] Moronis A X, Simou N, Kiousis K N and Fylladitakis E D 2014 *IEEE Trans. Dielectr. Electr. Insul.* **21** 1035–43
[18] Zhang J, Kong L, Qu J, Wang S and Qu Z 2019 *Energy* **171** 624–30
[19] Zhang J and Lai F C 2011 *J. Electrost.* **69** 486–93
[20] Wang J, Zhu T, Cai Y-X, Wang J-B and Wang J 2020 *Appl. Therm. Eng.* **169** 114922
[21] Feng J, Wang C, Liu Q and Wu C 2019 *Int. J. Heat Mass Transfer* **130** 640–9
[22] Pekárek S 2014 *Plasma Sources Sci. Technol.* **23** 062001
[23] Lv F, Song J, Wang P, Ruan H and Geng J 2019 *IEEE Access* **7** 123671–8
[24] Zhang Y, Li D and Wang H 2010 *Plasma Sources Sci. Technol.* **12** 702
[25] Zeng M J, Zhang J F, Wang S and Qu Z G 2021 *Appl. Therm. Eng.* **185** 116340
[26] Iranshahi K, Martynenko A and Defraeye T 2020 *Energy* **208** 118168
[27] Trichel G W 1938 *Phys. Rev.* **54** 1078–84
[28] Loeb L B, Kip A F, Hudson G G and Bennett W H 1941 *Phys. Rev.* **60** 714–22
[29] Lama W L and Gallo C F 1973 *J. Phys. D: Appl. Phys.* **6** 1963–72
[30] Lama W L and Gallo C F 1974 *J. Phys. D: Appl. Phys.* **45** 103–13
[31] Abdel-Salam M and Mufti A 1998 *Electr. Power Syst. Res.* **44** 145–54
[32] Abdel-Salam M, Farghally M and Abdel-Sattar S 1983 *IEEE Trans. Electr. Insul.* **EI–18** 110–9
[33] Morrow R 1985 *Phys. Rev.* A **32** 1799
[34] Chen J and Davidson J H 2003 *Plasma Chem. Plasma Process.* **23** 83–102
[35] Napartovich A P, Akishev Y S, Deryugin A A, Kochetov I V, Pan'kin M V and Trushkin N I 1997 *J. Phys. D: Appl. Phys.* **30** 2726
[36] Shaygani A and Adamiak K 2023 *J. Appl. Phys.* **56** 385202
[37] Hagelaar G J M and Pitchford L C 2005 *Plasma Sources Sci. Technol.* **14** 722
[38] Moss M S, Yanallah K, Allen R W K and Pontiga F 2017 *Plasma Sources Sci. Technol.* **26** 035009
[39] Calvo E M, Pinheiro M J and Sá P A 2022 *Appl. Sci.* **12** 1637
[40] Feng N, Ma T, Chen C, Yao B and Gao W 2022 *Energies* **15** 6431
[41] Chung J H, Sohn D K and Ko H S 2023 *Micromachines* **14** 1614
[42] Adamiak K, Atrazhev V and Atten P 2005 *IEEE Trans. Dielectr. Electr. Insul.* **12** 1015
[43] Chuan L, Zhi L, Pengyu W, Ming Z, Yong Y and Kexun Y 2020 *Plasma Sources Sci. Technol.* **29** 045011
[44] Adamiak K and Atten P 2004 *J. Electrost.* **61** 85
[45] Wang S, Qu J G, Kong L J, Zhang J F and Qu Z G 2019 *Int. J. Therm. Sci.* **139** 176
[46] Kulikovsky A A 1997 *J. Phys. D: Appl. Phys.* **30** 441
[47] Dordizadeh P, Adamiak K and Castle G P 2017 *J. Electrost.* **88** 49–54
[48] Sato N 1980 *J. Phys. D: Appl. Phys.* **13** L3
[49] Boeuf J P and Pitchford L C 2005 *J. Appl. Phys.* **97** 103307
[50] Deng F C, Ye L Y and Song K C 2013 *J. Phys. D: Appl. Phys.* **46** 425202
[51] Tran T N, Golosnoy I O, Lewin P L and Georghiou G E 2010 *J. Phys. D: Appl. Phys.* **44** 015203
[52] Sattari P, Gallo C F, Castle G S P and Adamiak K 2011 *J. Phys. D: Appl. Phys.* **44** 155502